

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 1.2 MECHANICAL PROPERTIES OF MATERIALS: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Imagine you are advising the farmer in Kericho whose sisal footbridge ropes behaved so differently under an 80 kg sack of tea leaves. Answer each question in the space under it, in your own words, using evidence from Lessons 1 to 6, your Summary Table, your force-extension graphs and your Final Model. Use correct vocabulary (elastic, plastic, elastic limit, spring constant, Hooke's Law, elastic potential energy, stress, strain, Young's Modulus). Show every calculation step with units, and explain WHY and HOW, not just WHAT. Take $g = 9.8 \text{ N/kg}$. You may add labelled diagrams. You have 80 to 100 minutes and you work individually.

Part 1: Elastic and Plastic Behaviour – What Did We Observe?

Remember the two sisal ropes on the Kericho suspension footbridge, and the three materials you stretched in Lessons 1 and 2 (rubber band, sisal twine, copper wire). (a) Describe what happened to each rope when the 80 kg sack was loaded and then removed. (b) Describe what you observed for each of the three materials in your stretching investigation. (c) Define elastic behaviour and plastic deformation, and use your observations to classify each material. (d) Explain why two ropes that look similar can behave so differently.

Write your answer here:

Part 2: What Is Happening Inside the Material?

Use a labelled particle-level diagram and a short explanation to show what happens to the particles and the bonds between them (a) when an elastic material is stretched and the force is removed, and (b) when a material is stretched beyond its elastic limit. Then say how your thinking about this has changed since the first model you sketched in Lesson 1.

Write your answer here:

Part 3: Hooke's Law, Elastic Limit and Stored Energy – Calculating for the Kericho Ropes

Use this data about the two ropes, with the load being the 80 kg tea-leaf sack.

Rope	Spring constant k (N/m)	Extension under the 80 kg load (m)
Rope A (new)	850	0.92
Rope B (old)	210	3.73

The elastic limit of Rope B is an extension of 2.1 m. (a) Calculate the weight of the sack. (b) State Hooke's Law and explain what the spring constant tells you about a material. (c) Calculate the force at Rope B's elastic limit and the mass this represents (its safe working load). (d) Calculate the elastic potential energy stored in Rope B at its elastic limit, and in Rope A at an extension of 0.92 m. (e) Using your answers, explain why Rope B failed and Rope A did not. Sketch the two force-extension graphs side by side and mark the load and the elastic limit of Rope B.

Write your answer here:

Part 4: Stress, Strain and Young's Modulus – Predicting Before Loading

(a) Define stress, strain and Young's Modulus, giving the equation and the unit (if any) for each. (b) Explain why a farmer or engineer should compare stress and strain rather than only force and extension when choosing a rope. (c) Name and describe the key features of a stress-strain graph that you identified in Lesson 5. (d) Explain how measuring Young's Modulus and fracture stress BEFORE loading would have let the farmer predict which rope would snap.

Write your answer here:

Part 5: Final Explanation – Advice to the Kericho Farmer

Write a formal recommendation to the Kericho farmer that answers the driving question: Why do some materials stretch and snap back while others permanently deform or break, and how can we use this knowledge to choose the right material for a safe structure in Kenya? Your answer must (a) explain why the two ropes behaved differently, using the elastic limit, spring constant and energy ideas; (b) give a safe working load for the old rope and say what you recommend he does; (c) describe one other Kenyan application, such as a matatu suspension spring in Nairobi or a bungee cord at a Nairobi adventure park, where the same ideas help choose a safe material; and (d) explain how your understanding has grown from your first prediction in Lesson 1.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of mechanical properties	Defines elastic, plastic, elastic limit, spring constant, stress, strain and Young's Modulus accurately, and links them to a particle-level explanation of bonds.	Defines most terms correctly with a minor gap or imprecision in the particle-level explanation.	Definitions are incomplete or confused, for example mixing up elastic limit and breaking point, and there is little or no particle-level explanation.
Use of evidence	Uses specific data from the lessons accurately (the 100–500 g stretching results, k values of 850 and 210 N/m, extensions of 0.92 m and 3.73 m, elastic limit 2.1 m).	Uses some specific lesson evidence, with a few missing or inexact values.	Uses general statements with little or no specific data from the investigations.
Scientific reasoning and calculations	Calculations (weight, safe working load, E_p) are correct, with steps and units shown, and the reasoning links the load to the elastic limit to the failure.	Calculations are mostly correct with small errors in units or steps, and the reasoning is mostly complete.	Calculations have major errors or are missing, and the reasoning is a list of facts that does not explain why the rope snapped.
Application to Kenyan contexts	Gives a clear, practical recommendation to the farmer and explains a second Kenyan application (matatu suspension spring or bungee cord) correctly.	Gives a reasonable recommendation and mentions a second application with limited explanation.	Recommendation is vague or unsupported, and there is no clear second application.

Communication and growth	Writing is clear, organised and uses correct vocabulary and labelled diagrams, and shows a clear journey from the Lesson 1 prediction to evidence-based explanation.	Writing is mostly clear and organised, with some growth from Lesson 1 described.	Writing is hard to follow, with little vocabulary use and no description of how understanding changed.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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